

MODELLING BLOOD FLOW IN THE CIRCULATORY SYSTEM

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ABSTRACT

Blood plays a crucial role in the human body as it delivers oxygen, nutrients and other essential elements to tissues and organs throughout the body. This project will model blood flow through vessels by reducing the Navier-Stokes equations to a cylindrical geometry. The blood vessels are first modelled as rigid elliptical tubes to derive the Hagen-Poiseuille flow, this can then be made more specific for circular tubes. The effects of smooth blockages are then explored, looking at the impact it has on different flow profiles and investigating the speed and pressure differences in these constrained domains. Next, smaller vessels are considered, using lubrication theory to rescale the momentum equations. Finally, pulsatile flow is introduced through an oscillatory pressure gradient, leading to the Womersley number.

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1 Introduction

1.1 Background

Blood flow is essential for transporting oxygen and nutrients throughout the body, allowing every organ to function properly. The behaviour of this flow depends on several factors including vessel geometry, blood viscosity and whether the flow is steady or pulsatile. These factors contribute to key quantities such as velocity, pressure, flow rate and wall shear stress.

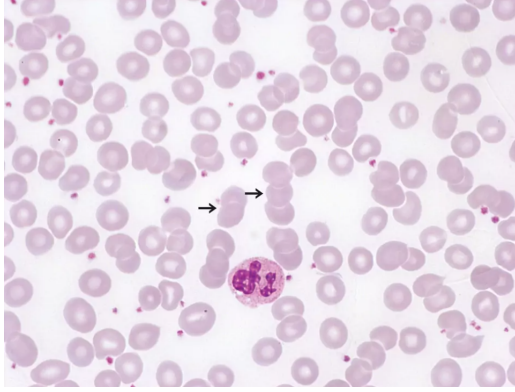
The movement of blood within the vessels can be modelled using the Navier-Stokes equations and continuity equation. This project aims to derive reduced forms of these governing equations under appropriate assumptions for blood flow. Starting with Hagen-Poiseuille flow, as a baseline model, it is then extended to consider the effect of blockages, flow in capillaries and pulsatile flow described by the Womersley model. Finally, these models are extended to consider the presence of viscoelastic properties and time-dependent boundaries.

1.2 Newtonian vs Non-Newtonian Fluid

Blood will be modelled as a Newtonian fluid, which means that the viscosity is constant regardless of shear forces. This is an appropriate and realistic assumption to make as the blood behaves as a Newtonian fluid in larger blood vessels such as the aorta near the heart. In the aorta, the diameter is relatively large in comparison to the size of red blood cells (RBCs), resulting in higher shear rates. This causes the viscosity to be independent of shear rate, justifying the Newtonian assumption.

Blood can also be modelled as a homogeneous suspension, as approximately 45% of the fluid consists of RBCs, which are the primary factor determining the viscosity of the blood. Suspensions containing rigid spherical particles behave approximately as Newtonian fluids when the particle concentration is small, as the effective viscosity remains independent of the shear rate. Unlike rigid particles, healthy RBCs are deformable and will change shape when subjected to shear stress in a flowing fluid. The rate of deformation depends on the shear rate and the concentration of the cells within the suspension.

In general, blood is modelled to be non-Newtonian and as a shear thinning fluid, meaning that the apparent viscosity decreases as the shear rate increases. This behaviour of the blood is typically found in smaller blood vessels such as the capillaries. At low shear rates, RBCs aggregate into *rouleaux* (as seen in Figure 1.1) which increases the viscosity of the blood. When shear rate reaches 1000 s^{-1} , the non-Newtonian behaviour becomes negligible causing the viscosity to become constant.



(a) Microscopic view of rouleaux (coin stacking) phenomenon [8].



(b) Visualisation of the rouleaux effect on the red blood cells [21].

Figure 1.1: Rouleaux (coin stacking) of red blood cells affecting the viscosity.

1.3 Structure of Blood Vessels

The blood vessels are described to be elastic, which means that they can deform due to forces being applied to them but can return to their original shape without dissipation of energy. The fibres that contribute to the elastic properties are elastin, collagen and smooth muscle.

1.3.1 Structure of Aorta

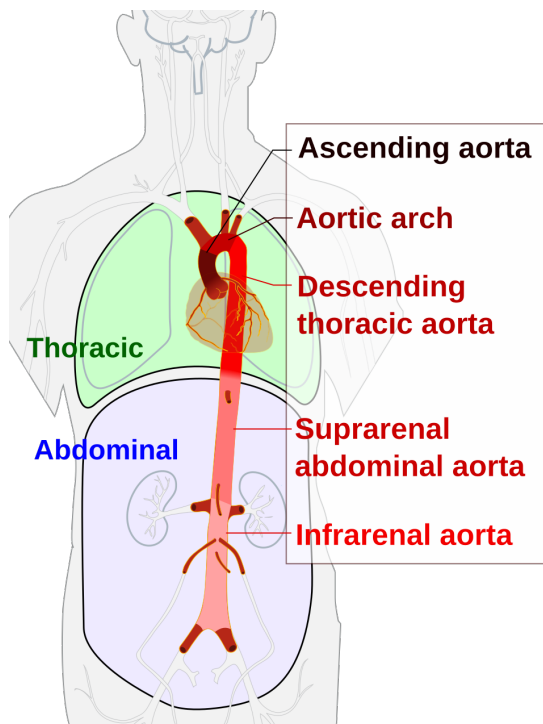
Large arteries, such as the aorta, have a wall made up of three layers: the *tunica intima*, *tunica media* and *tunica adventitia*. The tunica intima forms the inner lining of the vessel and consists of a single layer of endothelial cells that extends continuously throughout the circulation. In large central arteries like the aorta, the tunica media contains multiple concentric layers of elastic tissue separated by connective tissue, collagen fibres and smooth muscle cells. These elastic layers allow the vessel to stretch and recoil as blood is pumped from the heart. The location of the aorta in the abdomen and in connection to the heart are shown in Figure 1.2 and the full structure of the blood vessels is shown in Figure 1.3.

1.3.2 Structure of Capillaries

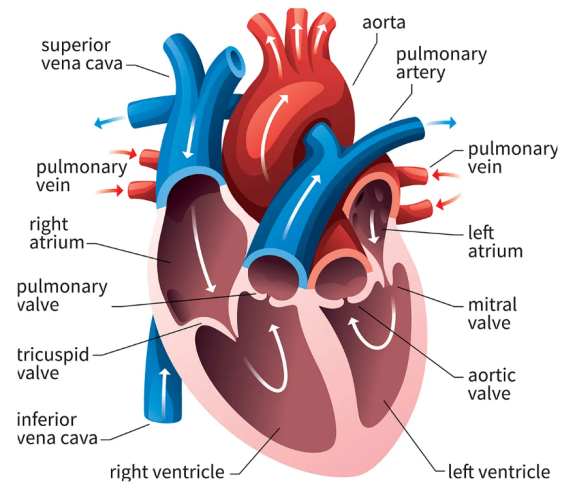
Capillaries are the smallest blood vessels in the circulatory system and have a simpler structure. The capillary wall consists of a single layer of endothelial cells supported by a basement membrane which may enclose occasional cells called *pericytes* [6] as seen in Figure 1.4. Unlike larger vessels, capillaries do not contain smooth muscle cells in their walls. Their walls are extremely thin, with a typical thickness of around $0.7 \mu\text{m}$.

2 Hagen-Poiseuille Flow

In this chapter, relevant and suitable assumptions will be made in order to reduce the Navier-Stokes equations to a form that can help describe flow through cylindrical pipes to model flow through blood vessels. This particular application of the Navier-Stokes equations to a cylindrical geometry is known as the *Hagen-Poiseuille flow*.

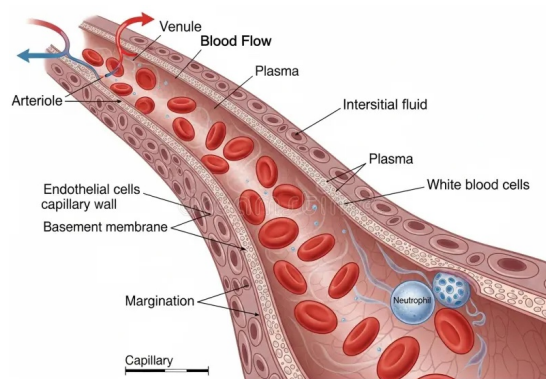


(a) Location of the heart in the abdomen along with the blood vessels emerging from the heart [24].

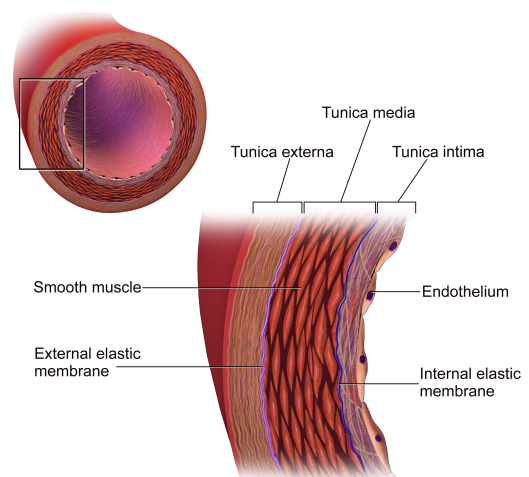


(b) Anatomy of the heart [11].

Figure 1.2: Location and anatomy of the human heart.



(a) General structure of a blood vessel [20].



(b) Cross section of blood vessel walls [26].

Figure 1.3: Structure of blood vessels and the cross section of larger blood vessels.

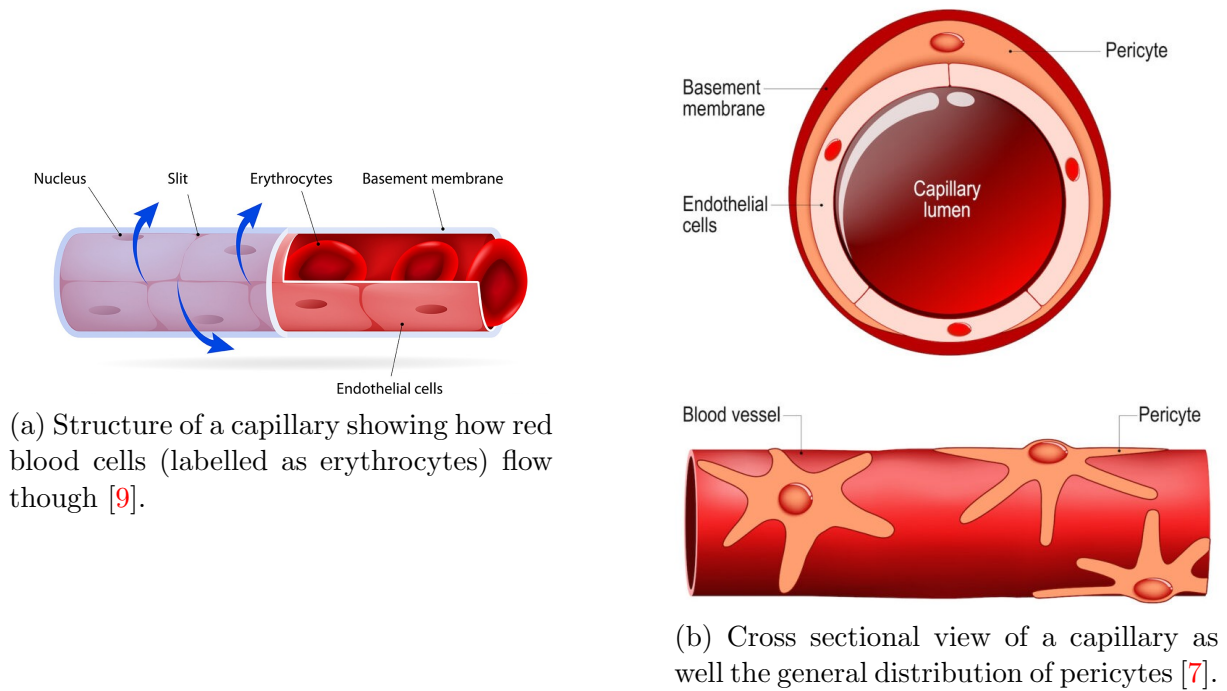


Figure 1.4: Structure of a human capillary.

2.1 Blood Flow in an Elliptical Tube

The Hagen-Poiseuille flow describes steady, laminar flow of a viscous incompressible fluid through a cylindrical pipe whose length is long compared to its diameter. It also provides a relationship between the pressure drop and the resulting flow. This model can be used to approximate blood flow in the aorta due to the similarities between its assumptions and the characteristics of blood flow in large vessels.

When modelling the aorta as a cylindrical tube, this provides a useful approximation for flow in large vessels. The specific aorta considered is the ascending aorta, which is connected to the left ventricle of the heart as seen Figure 1.2a. When the pressure in the left ventricle is higher than the pressure in the aorta (seen in Figure 1.2b), the valve opens and blood flows into the aorta. As the heart contracts, blood is ejected from the left ventricle into the aorta.

2.2 Assumptions for the Hagen-Poiseuille Flow

In cylindrical polar coordinates, the Navier-Stokes equations for the velocity profile $\mathbf{u} = u\hat{\mathbf{r}} + v\hat{\boldsymbol{\theta}} + w\hat{\mathbf{z}}$ can be written as

$$\begin{aligned}\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial r} + \frac{v}{r} \frac{\partial u}{\partial \theta} + w \frac{\partial u}{\partial z} - \frac{v^2}{r} \right) &= -\frac{\partial p}{\partial r} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2} - \frac{u}{r^2} - \frac{2}{r^2} \frac{\partial v}{\partial \theta} \right] \\ \rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial r} + \frac{v}{r} \frac{\partial v}{\partial \theta} + w \frac{\partial v}{\partial z} + \frac{uv}{r} \right) &= -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v}{\partial \theta^2} + \frac{\partial^2 v}{\partial z^2} + \frac{2}{r^2} \frac{\partial u}{\partial \theta} - \frac{v}{r^2} \right] \\ \rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial r} + \frac{v}{r} \frac{\partial w}{\partial \theta} + w \frac{\partial w}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial w}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 w}{\partial \theta^2} + \frac{\partial^2 w}{\partial z^2} \right]\end{aligned}$$

For a cylinder of radius a , the Navier-Stokes equations can be reduced by using the following assumptions:

- no-slip boundary at the wall: $u = v = w = 0$ at $r = a$,
- steady flow: $\frac{\partial \mathbf{u}}{\partial t} = \frac{\partial p}{\partial t} = 0$,
- fully developed flow: $\frac{\partial w}{\partial z} = 0$,
- axisymmetric flow: $\frac{\partial \mathbf{u}}{\partial \theta} = 0$,
- no swirl, $v = 0$,
- constant axial pressure gradient: $\frac{\partial p}{\partial z} = G$ (constant).

Since the flow only has an axial component, then $\mathbf{u} = (0, 0, w(r))$.

2.3 Elliptical Derivation of the Hagen-Poiseuille Flow

Blood vessels are not perfectly circular, and their cross-sections can take on other shapes. As a result, the Hagen-Poiseuille flow is derived here for an elliptical cross-section with the circular case being recovered as a special case.

The flow is considered in a straight vessel aligned with the z -direction. The cross-section is taken to be elliptical, with the semi-axes a and b in the x - and y -directions, respectively, and $a > b$ without loss of generality (the general blood vessel schematic can be seen in Figure 2.1).

The boundary of the vessel is therefore given by:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

For circular pipes, the velocity depends only on the radial component. However, for

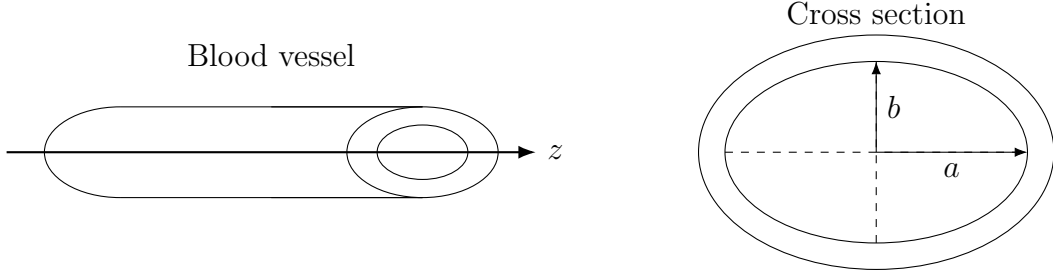


Figure 2.1: Schematic representation of the blood vessel cross-section showing the minor and major radii.

an elliptical cross-section, there is no radial symmetry resulting in the velocity being a function of both x and y .

The velocity profile of this flow is in the form:

$$w(x, y) = c \left(1 - \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) \right). \quad (2.1)$$

where c is a constant that represents the maximum velocity. This form is consistent with known analytical solutions for steady, unidirectional flow in tubes of elliptical cross-section [2].

The pressure gradient is assumed to be constant in the axial direction and is therefore expressed as:

$$\frac{\partial p}{\partial z} = -G \quad (2.2)$$

where $G > 0$ represents the magnitude of the change in pressure.

Substituting into the Navier-Stokes equations, the convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ vanishes since the flow is steady, fully developed and unidirectional. Therefore, the momentum equations reduce to

$$\frac{\partial p}{\partial z} = \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right). \quad (2.3)$$

Using Equation (2.2), this reduces to

$$-\frac{G}{\mu} = \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2}. \quad (2.4)$$

The right-hand side of Equation (2.4) can be found by differentiating Equation (2.1), so

$$\frac{\partial^2 w}{\partial x^2} = -2ca^{-2} \quad , \quad \frac{\partial^2 w}{\partial y^2} = -2cb^{-2}.$$

Substituting into Equation (2.4) gives

$$-\frac{G}{\mu} = -2ca^{-2} + -2cb^{-2} \quad (2.5)$$

Rearranging for c gives

$$c = \frac{G}{2(a^{-2} + b^{-2})\mu} = \frac{Ga^2b^2}{2\mu(a^2 + b^2)}. \quad (2.6)$$

The volumetric flow rate is given by

$$Q = \iint_A w(x, y) dx dy \quad (2.7)$$

where A is the elliptical cross-section defined by

$$A = \left\{ (x, y) : \frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1 \right\}. \quad (2.8)$$

Substituting Equation (2.1) into Equation (2.7) gives

$$Q = c \iint_A \left(1 - \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) \right) dx dy. \quad (2.9)$$

The flux can be calculated using the parametric substitutions $x = ak \cos \theta$, $y = bk \sin \theta$, where $0 \leq k \leq 1$ and $0 \leq \theta \leq 2\pi$. This means the area element becomes $dx dy = abk dk d\theta$.

$$\begin{aligned} Q &= c \int_0^{2\pi} \int_0^1 (1 - k^2) abk dk d\theta \\ &= 2\pi abc \int_0^1 k(1 - k^2) dk \\ &= 2\pi abc \left[\frac{k^2}{2} - \frac{k^4}{4} \right]_0^1 \\ &= \frac{\pi abc}{2}. \end{aligned}$$

Substituting Equation (2.6) into above gives the flow rate:

$$Q = \frac{G\pi a^3 b^3}{4\mu(a^2 + b^2)}. \quad (2.10)$$

Using the same substitution into Equation (2.1) gives the velocity:

$$w(x, y) = \frac{Ga^2b^2}{2\mu(a^2 + b^2)} \left(1 - \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) \right). \quad (2.11)$$

The volumetric flow rate Q describes the volume of fluid passing through the cross-section of the vessel per unit time.

2.4 Explorations

Above, the velocity is defined in Equation (2.1) and is zero at the boundary, due to the no-slip condition. As the radius gradually increases from the centre to the wall (0 to a), the velocity gradually decreases. As the fluid moves further away from the boundary, it reduces the resistance between the blood and the vessel wall, resulting in the maximum velocity being found in the centre of the blood vessel.

2.4.1 Maximum Speeds

In circular pipe flows, the central velocity is found to be twice the average velocity field, which is given by $\bar{w} = \frac{Q}{A}$ [6]. The relationship can be verified for an elliptical cross-section. Using Q from before and $A = \pi ab$, the average velocity is given by

$$\bar{w} = \frac{Ga^2b^2}{4\mu(a^2 + b^2)} \quad (2.12)$$

The central channel velocity can be found by setting $x = 0$ and $y = 0$ in w , then

$$w(0, 0) = w_{max} = \frac{Ga^2b^2}{2\mu(a^2 + b^2)},$$

confirming that $w_{max} = 2\bar{w}$.

Despite the structure of the blood vessel changing from a circular cross-sectional area to an elliptical cross-sectional one, the relationship between w_{max} and $2\bar{w}$ remains the same. The velocity profile is parabolic, meaning that it is the fastest in the centre and gradually decreases towards the boundary. Hence, w_{max} and $2\bar{w}$ still hold for an elliptical cross-section.

2.4.2 Velocity and Varying Eccentricity

It is easy to assume that the flow becomes more restricted as the eccentricity¹ of the vessel increases, but it is important to see how that arises. Let $a > b$, so the vessel is stretched in the x -direction and flattened in the y -direction (as seen in Figure 2.1). For example, $a = 1, b = 0.5$ results in $w(x, y) = c(1 - (x^2 + 4y^2))$. The change in the cross-sectional area means that the blood flows closer to the boundary in the y -direction than it does in

¹For an ellipse with semi-major axis $a > 0$ and semi-minor axis $b > 0$ (namely $a > b > 0$), the eccentricity $e \in [0, 1)$ of this ellipse is given by $e = \sqrt{1 - \frac{b^2}{a^2}}$, thus a circle ($a = b$) will have 0 eccentricity.

the x -direction. This results in the velocity decreasing faster in this direction due to the no-slip condition.

2.4.3 Streamlines

Streamlines describe the flow of the fluid at any given point in time and, for blood flows in an elliptical tube, they can be found by using the set of equations [23]:

$$\frac{dx}{ds} = 0, \quad \frac{dy}{ds} = 0, \quad \frac{dz}{ds} = w(x, y) = \frac{Ga^2b^2}{2\mu(a^2 + b^2)} \left(1 - \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} \right) \right),$$

where s is a parameter along the streamline. The first two differential equations imply that x and y are both constants, say $x = x_0$ and $y = y_0$. These can be used to find z as

$$z = c \left(1 - \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} \right) \right) s + z_0.$$

The streamlines obtained are constant and parallel to the z -axis. Given that the flow is steady, the streamlines are expected to be constant.

2.4.4 Flow Rate and Varying Eccentricity

The velocity profile depends on the eccentricity of the ellipse, this section examines how to flow rate is affected by such changes. For a circular cross-section of radius R , the volumetric flow rate is given by Equation (2.10) with $R = a = b$. If the radius doubles then

$$Q = \frac{2G\pi R^4}{\mu}.$$

This shows that doubling the radius size results in an increase in the flow rate by a factor of 16.

To allow a meaningful comparison, consider a circular cross-section with the same area as the elliptical cross-section. Take an ellipse with $b = \frac{a}{2}$, so its area is $A = \pi ab = \frac{\pi a^2}{2}$. A circle with equivalent area has the radius $R = \frac{a}{\sqrt{2}}$. Substituting this into the circular flow expression gives

$$Q_c = \frac{G\pi a^4}{32\mu}. \quad (2.13)$$

Comparing this with the elliptical case,

$$Q_e = \frac{Q\pi a^4}{40\mu}. \quad (2.14)$$

Hence, for equal cross-sectional area $Q_e = 0.8Q_c$, showing a 20% decrease in flow rate.

For increased eccentricity, this reduction becomes more pronounced. For example, tak-

ing $b = \frac{a}{3}$ and comparing with a circular cross-section of equal area gives $Q_e = 0.6Q_c$, a 40% decrease in flow rate. This shows how reduction in flow is not fixed but depends on the degree of geometric change.

If $b \ll a$, then the flow rate can be approximated as

$$Q \approx \frac{G\pi a^3 b^3}{4\mu a^2} \approx \frac{G\pi a b^3}{4\mu}.$$

Fixing $a = 1$ and varying b , the change in flow rate can be studied. For $b = 0.1, 0.01$ and 0.001 the flow rate is evaluated as $Q = 2.5 \times 10^{-4} \times \frac{G\pi}{\mu}$, $2.5 \times 10^{-7} \times \frac{G\pi}{\mu}$ and $2.5 \times 10^{-10} \times \frac{G\pi}{\mu}$. This shows that as $b \rightarrow 0$ then $Q \rightarrow 0$.

2.4.5 Wall Shear Stress

Wall shear stress (WSS) is the frictional force of the blood on the endothelial cells which can be found by multiplying fluid shear viscosity by the normal derivative of tangential velocity, or more formally:

$$\tau_w = \mu \frac{\partial w}{\partial n} = \mu \left(\frac{\partial w}{\partial x} n_x + \frac{\partial w}{\partial y} n_y \right), \quad (2.15)$$

with the elliptical normal being

$$\mathbf{n} = \frac{1}{\sqrt{\frac{4x^2}{a^4} + \frac{4y^2}{b^4}}} \left(\frac{2x}{a^2}, \frac{2y}{b^2} \right). \quad (2.16)$$

The WSS is found to be

$$\tau_w = -c\mu \sqrt{\frac{4x^2}{a^4} + \frac{4y^2}{b^4}},$$

where c is defined in Equation (2.6). This result is consistent with analyses of steady flow in elliptical areas [5].

For a circular cross-sectional area, set $a = b = R$, then the WSS is found to be

$$\tau_w = \frac{-2c\mu}{R}$$

which is a constant, meaning that for the circular structure, the blood has uniform WSS.

To examine whether WSS is uniform around the boundary, it can be evaluated at certain points. At the very top and bottom of the boundary, the points would be $(0, \pm b)$, resulting in

$$\tau_w = \frac{-2c\mu}{b},$$

and similarly, for the far left and right of the boundary, the points would be $(\pm a, 0)$ which

results in

$$\tau_w = \frac{-2c\mu}{a}.$$

Unlike before, the WSS is not uniform everywhere in the flow. If $a > b$, like before, then the shear stress is higher at $(0, \pm b)$, than at $(\pm a, 0)$.

Shear stress arises from the relative motion of adjacent layers of fluid, with the shear rate measuring how quickly the velocity changes [6]. Along the longer semi-axis from $(0, 0)$ to $(a, 0)$, the velocity decreases from its maximum value to zero over a larger distance, so neighbouring fluids move at a more similar speed and the velocity gradient, or shear rate, is smaller. In contrast, along the shorter semi-axis $(0, 0)$ to $(0, b)$, the same decrease in velocity occurs over a shorter distance. Therefore, the velocity changes more rapidly between adjacent layers, producing a larger shear rate and hence a larger WSS.

As the stress in the circulation increases, it causes RBCs to deform but given the elastic nature of the RBCs, they can return back to their original shape. On the other hand, with endothelial cells, when a shear stress of 40 Nm^{-2} is applied, it causes the cells to rupture and tear away [6].

2.5 Limitations of the Hagen-Poiseuille Model

The Hagen-Poiseuille model is useful for describing steady flow in larger vessels with the assumption of rigid vessel walls, Newtonian fluid and a vessel that is larger in comparison to the RBCs. However, this is not sufficient to model all vessels in the circulatory system. Chapter 3 extends this model to examine how blockages and constrictions affect flow rate, pressure and wall shear stress while using the inviscid Bernoulli equation.

3 Blockages and Constricted Flow

Blood flow in blood vessels is not always ideal. In particular, plaque buildup, the formation of blood clots and the narrowing of vessels can disrupt the normal flow of blood. These changes in vessel geometry can significantly affect quantities such as pressure, velocity, flow rate and wall shear stress.

3.1 Smooth Constrictions

3.1.1 Flow Through a Smooth Constriction

To model the blockage in a vessel, consider a small rigid cylindrical tube where the cross-sectional area decreases from A_1 to a smaller area, A_2 and gradually increases back to A_1 . The region of cross-sectional area A_2 represents the area in the vessel affected by the blockage. This particular geometry is known as the *Venturi section* and is shown in Figure 3.1.

Let U_1 and U_2 denote the average velocities upstream and at the constricted area. Assuming incompressible flow and the flow rate is the same in both sections:

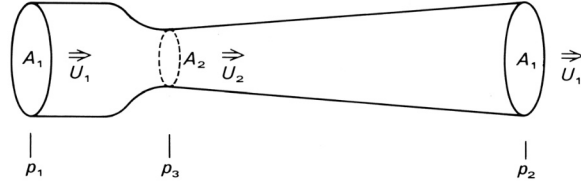


Figure 3.1: Flow in a Venturi section where initially the tube cross-sectional area is decreased from A_1 to A_2 , and then slowly re-expands to area A_1 . The pressure measurements of p_1, p_2 and p_3 are made upstream, downstream and at the of the location of the constriction [6].

$$Q = U_1 A_1 = U_2 A_2. \quad (3.1)$$

Since $A_1 > A_2$ then $U_1 < U_2$, so the fluid accelerates as it goes through the constricted area. If viscous forces are neglected, Bernoulli's equation $p + \frac{1}{2}\rho u^2 = \text{constant}$ can be applied across the section where the area changes. Using Figure 3.1, this gives

$$p_1 - p_3 = \frac{1}{2}\rho(U_2^2 - U_1^2). \quad (3.2)$$

As the flow accelerates through the constriction, the increase in velocity is accompanied by a decrease in pressure. Using $U_2 = U_1 \frac{A_1}{A_2}$ with $Q = U_1 A_1 = U_2 A_2$ gives

$$Q = \sqrt{\frac{2\Delta p}{\rho\left(\frac{1}{A_2^2} - \frac{1}{A_1^2}\right)}}, \quad (3.3)$$

where $\Delta p = p_1 - p_3$.

3.1.2 Blockage Effect on Flow Rate

As discussed in Chapter 2, the flow rate is highly sensitive to changes in the vessel geometry, which was seen when varying the eccentricity of the cross-section. This resulted in large changes to Q . A similar effect occurs when a blockage reduces the vessel radius, restricting the flow.

For a circular vessel, the Hagen-Poiseuille flow gives

$$Q = \frac{GR^4\pi}{8\mu}. \quad (3.4)$$

Reducing the radius by a small amount leads to a larger reduction in flow rate. The relationship between the flow rate Q and radius R can be expressed as $Q \propto R^4$ thus:

$$\frac{Q_r}{Q_o} = \left(\frac{R_r}{R_o}\right)^4 \quad (3.5)$$

where the subscript r denotes the reduced radius system and the subscript o denotes the original system. To observe what happens when the flow rate reduces by 10%, set $\frac{Q_r}{Q_o} = 0.9$ which gives $\frac{R_r}{R_o} \approx 0.974$.

A 2.6% reduction in the radius is required to reduce the flow rate by 10%. Reducing the flow rate to half of its original value requires only a 16% reduction in the radius, illustrating that relatively small blockages can have a significant effect on blood flow.

3.2 Hagen-Poiseuille Revisited

In this section, the Hagen-Poiseuille flow is revisited to explore the impact of blockages on key flow quantities. The velocity of this flow is given by [22]

$$w(r) = -\frac{1}{4\mu} \frac{dp}{dz} (a^2 - r^2). \quad (3.6)$$

3.2.1 Blockage Effect on Wall Shear Stress

The wall shear stress is defined by

$$\tau_w = -\mu \frac{\partial w}{\partial r} \quad (3.7)$$

which evaluates to

$$\tau_w = -\frac{\partial p}{\partial z} \frac{R}{2} \quad (3.8)$$

Using the relation $Q = \frac{G\pi R^4}{8\mu}$, where $G = -\frac{\partial p}{\partial z}$, the pressure gradient can be written as

$$-\frac{\partial p}{\partial z} = \frac{8\mu Q}{\pi R^4}. \quad (3.9)$$

Substituting into the expression τ_w gives

$$\tau_w = \frac{4\mu Q}{\pi R^3} \quad (3.10)$$

To model a blockage, let the radius vary with axial position, so that $R = R(z)$. Let z_1 denote a location upstream and z_2 a location at the constriction so that $R(z_1) > R(z_2)$. Then

$$\tau_w(z_1) = \frac{4\mu Q}{\pi(R(z_1))^3} < \tau_w(z_2) = \frac{4\mu Q}{\pi(R(z_2))^3}.$$

Therefore, the wall shear stress increases in the constricted region.

3.2.2 Blockage Effect on the Pressure Gradient

The pressure gradient is given by Equation (3.9). Allowing the radius to vary, this becomes

$$-\frac{\partial p}{\partial z} = \frac{8\mu Q}{\pi(R(z))^4} \quad (3.11)$$

Since $R(z_1) > R(z_2)$ it follows that

$$\frac{8\mu Q}{\pi(R(z_1))^4} < \frac{8\mu Q}{\pi(R(z_2))^4}$$

From Bernoulli's equation, $\Delta p = p_1 - p_3$ and extending to this new model $\Delta p = p(z_1) - p(z_2)$. Using the fundamental theorem of calculus, a formula can be obtained for change in pressure as

$$\Delta p = p(z_1) - p(z_2) = \int_{z_1}^{z_2} \left(-\frac{\partial p}{\partial z} \right) dz = \int_{z_1}^{z_2} \frac{8\mu Q}{\pi R(z)^4} dz \quad (3.12)$$

This shows that the pressure drop depends on the variation of the radius along the vessel.

3.2.3 Blockage Effect on Velocity

The velocity profile with varying radius is obtained by substituting Equation (3.9) into the Poiseuille solution, giving

$$w(r, z) = \frac{2Q}{\pi R(z)^4} (R(z)^2 - r^2) \quad (3.13)$$

The maximum velocity occurs at $r = 0$ giving

$$w(0, z) = w_{max}(z) = \frac{2Q}{\pi R(z)^2}.$$

Since $R(z_1) > R(z_2)$, it follows that $w_{max}(z_1) < w_{max}(z_2)$. Hence, the velocity increases in the constricted region.

Overall, results show that constrictions can have a significant impact on blood flow. As the available cross-sectional area decreases, both the velocity and wall shear stress increase while the pressure correspondingly decreases. In addition, the strong dependence of flow rate on the vessel radius highlights that even small reductions in radius can lead to substantial changes in flow. These findings demonstrate that obstructions and restrictions can have severe physiological consequences, as they can disrupt the transport of oxygen throughout the body.

3.3 Limitations of the Constricted Model

This chapter looks at how blockages and constrictions affect the steady flow in a rigid cylindrical tube. The use of Bernoulli's equation showed how minor changes to the geometric structure affected different flow quantities. However, this does not consider how the viscous behaviour of blood changes when flowing through a constricted area.

The next chapter looks at flow in much smaller geometries such as the capillaries. Here, the Reynolds number is very small and viscous forces dominate over inertial forces. It also rescales the momentum equations to derive the lubrication approximation for capillary

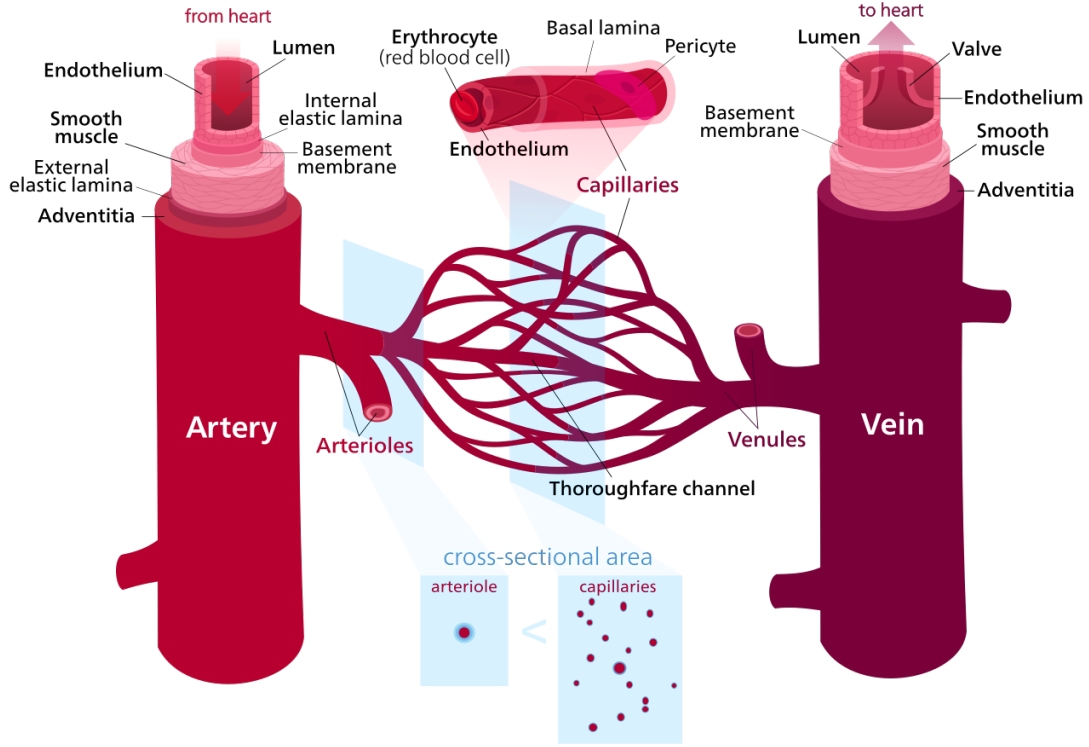


Figure 4.1: Structure of how arteries branch into arterioles before branching even further into capillaries. These branches then merge into venules which merge into veins [25].

flow.

4 Blood Flow in Capillaries

In the circulatory system, aortas branch out into arteries, then into the arterioles, and finally the capillaries, as seen in Figure 4.1. The capillaries are relatively smaller in size compared to the aortas, resulting in a change in blood flow behaviour. The combination of the blood vessels splitting into more vessels with a smaller diameter causes the velocity of the fluid to decrease since the total cross-sectional area of the vessels increases. As mentioned above, the viscosity in the capillaries increases (or is higher than in the aorta) due to lower shear rates and the aggregation of RBCs into rouleaux as seen in Figure 1.1.

4.1 Low Reynolds Number

The Reynolds number is a quantity that determines whether a fluid is dominated by either inertial forces or viscous forces. In this context, it is given by

$$Re = \frac{\rho U d}{\mu}$$

where ρ is the density, U is the characteristic velocity, d is the diameter of the blood vessel and μ is the dynamic viscosity. At sufficiently low Reynolds numbers, the influence of fluid inertia may be neglected. This is known as the *creeping flow approximation* and

is the relevant regime for blood flow in capillaries.

In this section, the governing equations are reduced by rescaling variables. As discussed in [15], when viscous flow only mildly deviates from ideal parallel walls, the character of the exact solution persists and the equations reduce to a set of lubrication equations.

Given a capillary of length L with radius a , the flow is assumed to be axisymmetric with no swirl. Under these assumptions, the continuity and Navier-Stokes equations reduce to:

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial r} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial w}{\partial r} \right) + \frac{\partial^2 w}{\partial z^2} \right], \quad (4.1)$$

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial r} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial p}{\partial r} + \mu \left[\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} - \frac{u}{r^2} + \frac{\partial^2 u}{\partial z^2} \right] \quad (4.2)$$

$$\frac{1}{r} \frac{\partial}{\partial r} (ru) + \frac{\partial w}{\partial z} = 0 \quad (4.3)$$

Let $\varepsilon = \frac{a}{L}$ be the ratio between the radius and length of the vessel. Given that $a \ll L$ then $0 < \varepsilon \ll 1$. The non-dimensional variables are given by:

$$t^* = \frac{Ut}{L}, \quad p^* = \frac{p}{P_a}, \quad w^* = \frac{w}{U}, \quad u^* = \frac{u}{\varepsilon U}, \quad r^* = \frac{r}{a} = \frac{r}{\varepsilon L}, \quad z^* = \frac{z}{L} \quad (4.4)$$

where P_a is the atmospheric pressure. Substituting these into the continuity and momentum equations gives:

$$\frac{1}{r^*} \frac{\partial}{\partial r^*} (r^* u^*) + \frac{\partial w^*}{\partial z^*} = 0 \quad (4.5)$$

$$\frac{\rho U^2}{L} \left(\frac{\partial w^*}{\partial t^*} + u^* \frac{\partial w^*}{\partial r^*} + w^* \frac{\partial w^*}{\partial z^*} \right) = -\frac{P_a}{L} \frac{\partial p^*}{\partial z^*} + \frac{\mu U}{\varepsilon^2 L^2} \left[\frac{1}{r^*} \frac{\partial}{\partial r^*} \left(r^* \frac{\partial w^*}{\partial r^*} \right) + \varepsilon^2 \frac{\partial^2 w^*}{\partial z^{*2}} \right], \quad (4.6)$$

$$\frac{\rho U^2 \varepsilon}{L} \left(\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial r^*} + w^* \frac{\partial u^*}{\partial z^*} \right) = -\frac{P_a}{\varepsilon L} \frac{\partial p^*}{\partial r^*} + \frac{\mu U}{\varepsilon L^2} \left[\frac{\partial^2 u^*}{\partial r^{*2}} + \frac{1}{r^*} \frac{\partial u^*}{\partial r^*} - \frac{u^*}{r^{*2}} + \varepsilon^2 \frac{\partial^2 u^*}{\partial z^{*2}} \right] \quad (4.7)$$

Multiplying Equation (4.6) by $\frac{\varepsilon^2 L^2}{\mu U}$ and Equation (4.7) by $\frac{\varepsilon L^2}{\mu U}$ gives the following:

$$\varepsilon^2 Re_L \left(\frac{\partial w^*}{\partial t^*} + u^* \frac{\partial w^*}{\partial r^*} + w^* \frac{\partial w^*}{\partial z^*} \right) = -\frac{1}{\Lambda} \frac{\partial p^*}{\partial z^*} + \frac{1}{r^*} \frac{\partial}{\partial r^*} \left(r^* \frac{\partial w^*}{\partial r^*} \right) + \varepsilon^2 \frac{\partial^2 w^*}{\partial z^{*2}}, \quad (4.8)$$

$$\varepsilon^2 Re_L \left(\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial r^*} + w^* \frac{\partial u^*}{\partial z^*} \right) = -\frac{1}{\Lambda} \frac{\partial p^*}{\partial r^*} + \frac{\partial^2 u^*}{\partial r^{*2}} + \frac{1}{r^*} \frac{\partial u^*}{\partial r^*} - \frac{u^*}{r^{*2}} + \varepsilon^2 \frac{\partial^2 u^*}{\partial z^{*2}}, \quad (4.9)$$

where $Re_L = \frac{\rho UL}{\mu}$ is the Reynolds number based on axial length and $\Lambda = \frac{\mu U}{P_a L}$ is a dimensionless parameter comparing viscous and pressure forces.

The resulting equations contain terms of order ε^2 and $\varepsilon^2 Re_L$. As $\varepsilon \ll 1$, these terms are negligible. Therefore, the inertial terms in Equation (4.8) are neglected and the axial viscous terms, which is of order ε^2 disappear, giving:

$$0 = -\frac{1}{\Lambda} \frac{\partial p^*}{\partial z^*} + \frac{1}{r^*} \frac{\partial}{\partial r^*} \left(r^* \frac{\partial w^*}{\partial r^*} \right), \quad (4.10)$$

Similarly, in Equation (4.9) the inertial and axial viscous terms are neglected, while the radial viscous terms remain. This gives:

$$0 = -\frac{\partial p^*}{\partial r^*} \quad (4.11)$$

These two equations can be used to obtain the flow rate which is

$$w^*(r) = \frac{1}{4\Lambda} \frac{\partial p^*}{\partial z^*} (r^{*2} - 1), \quad (4.12)$$

where $\frac{\partial p^*}{\partial z^*}$ is a constant, since pressure is independent of the radial component. This is the same dimensionless velocity profile obtained for the circular Hagen-Poiseuille flow.

Although the final expression is the same as the circular Hagen-Poiseuille flow, it arises because of different reasons. Before, the parabolic profile came from assuming steady flow in a cylindrical pipe. Here, the same profile comes from the Reynolds number being very small and the capillary being long and slender resulting in the inertial and axial viscous terms becoming negligible.

This does not mean that Poiseuille flow is only valid in capillaries but rather it shows that it is consistent with creeping flow. However, the flow cannot be the same in capillaries and arteries; the main limitation of using Hagen-Poiseuille flow to describe the flow in arteries is the assumption of steady flow. This neglects the pulsating nature of blood flows in large arteries.

4.2 Realistic Values of Re and ε

4.2.1 Capillaries

Using typical vessel dimensions, the Reynolds number and the aspect ratio can be found, helping to verify whether the assumptions above are reasonable. When $Re \ll 1$, viscous forces dominates over inertial forces, which is the regime creeping flow applies.

In capillaries, a typical diameter is $5.5 \mu\text{m}$, the blood velocity is around $700 \mu\text{m s}^{-1}$ and the length is approximately $615 \mu\text{m}$ [16]. Taking $\rho = 1.06 \text{ g cm}^{-3}$ and $\mu = 4 \times 10^{-3} \text{ Pa s}$ gives $Re = 0.001$. Since $Re \ll 1$, the inertial effects are negligible in capillary flow and the creeping flow approximation is well-justified. In addition, the aspect ratio can also be calculated. Taking the radius as $2.75 \mu\text{m}$ and the vessel length from before, $\varepsilon = 4.5 \times 10^{-3}$ giving $\varepsilon \ll 1$. This supports the neglect of the axial viscous effects relative to radial viscous effects in the reduced equations.

The calculations done show that the assumptions made in Section 4.1 are consistent with the physical scale of blood flow in capillaries. In particular, the very small Reynolds number confirms that viscous forces dominate, while the small aspect ratio supports the

lubrication-type approximation used to simplify the governing equations.

4.2.2 Arteries

The same can be done for the flow in aorta, but this demonstrates how creeping flow does not occur here. For the root of the aorta, the diameter is approximately 2.5 cm, the peak velocity during systole of 70 cm s^{-1} [16], density $\rho = 1.06 \text{ g cm}^{-3}$ and $\mu = 4 \times 10^{-3} \text{ Pa s}$, giving $Re = 4637.5$. This value is far greater than 1, so the inertial effects are important and creeping flow is not acceptable here.

However, in the ascending aorta, which is slightly further from the heart, has mean velocity 20 cm s^{-1} [16], giving $Re = 1325$. Despite it being less than the value at the root, it is still far greater than 1, meaning inertia still dominates over viscosity. Therefore, while creeping flow is appropriate in capillaries, it is evident that it is not for larger vessels such as the aorta. In these arteries, inertial effects become important due to the pumping action of the heart.

4.3 Limitations of the Creeping Flow Approximation

This chapter explored how flow behaves in much smaller geometries compared to the aorta. Although the velocity profile remains the same as in Chapter 2, it arises for a different reason: the low Reynolds number and slender geometry. Furthermore, this chapter introduced the Reynolds number, which was shown to differ significantly between capillaries and the aorta. This demonstrates that even though the velocity profile is the same, different forces dominate in different parts of the circulatory system.

However, what the previous chapters fail to capture is the pulsating factor in blood vessels. This introduces a time-dependent pressure gradient, which leads to unsteady flow, which is the main difference between Hagen-Poiseuille flow and creeping flow to pulsating flow. The next chapter addresses this by introducing an oscillatory pressure gradient and deriving the Womersley number, α .

5 Pulsating Blood Flow

The Hagen-Poiseuille model assumes steady flow. However, in large arteries the pressure gradient varies with time due to the pumping action of the heart. Introducing an oscillatory pressure gradient leads to the *Womersley flow*, which describes the velocity of the fluid in a rigid cylindrical vessel under pulsating forces. In the aorta, the pulsating force is due to the left ventricle in the heart having high amounts of blood causing a pressure difference between the left ventricle and the aorta. This difference causes the valve to open and blood to flow into the aorta. The resulting pulsating pressure gradient is the focus of the Womersley model.

5.1 Assumptions for Womersley Flow

The Womersley flow assumes the same conditions as Hagen-Poiseuille flow (axisymmetric, no swirl and fully developed), but with an oscillating axial pressure gradient of the form

$$\frac{\partial p}{\partial z} = -G(t). \quad (5.1)$$

5.2 Derivation of Womersley Number

The Womersley number comes from the non-dimensionalisation of the axial momentum equation for pulsatile flow [19], or equivalently by comparing the transient inertial and viscous forces [10]. Following the non-dimensionalisation approach, under the assumptions in Section 5.1, the Navier-Stokes equations reduce to:

$$\rho \frac{\partial w}{\partial t} = -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial r^2} + \frac{1}{r} \frac{\partial w}{\partial r} \right). \quad (5.2)$$

To non-dimensionalise this equation, the following variables are introduced:

$$r^* = \frac{r}{R}, \quad t^* = \frac{t}{T} = \omega t, \quad w^* = \frac{w}{W}.$$

where ω is the angular frequency of the pulsation. Substituting these into Equation (5.2) gives:

$$\rho \omega W \frac{\partial w^*}{\partial t^*} = -\frac{\partial p}{\partial z} + \mu \frac{W}{R^2} \left(\frac{\partial^2 w^*}{\partial r^{*2}} + \frac{1}{r^*} \frac{\partial w^*}{\partial r^*} \right). \quad (5.3)$$

Multiplying through by $\frac{R^2}{W\mu}$ gives

$$\frac{\rho \omega R^2}{\mu} \frac{\partial w^*}{\partial t^*} = -\frac{R^2}{W\mu} \frac{\partial p}{\partial z} + \left(\frac{\partial^2 w^*}{\partial r^{*2}} + \frac{1}{r^*} \frac{\partial w^*}{\partial r^*} \right). \quad (5.4)$$

The dimensionless group on the LHS of Equation (5.4) is denoted α^2 , where α is the Womersley number:

$$\alpha^2 = \frac{\rho \omega R^2}{\mu} = \frac{\omega R^2}{\nu} \quad (5.5)$$

where $\nu = \frac{\mu}{\rho}$ is the kinematic viscosity. Hence,

$$\alpha = R \sqrt{\frac{\omega}{\nu}}. \quad (5.6)$$

5.3 Womersley Number

5.3.1 Definition of the Womersley Number

The Womersley number $\alpha = R\sqrt{\frac{\omega}{\nu}}$, where ω is the angular frequency of oscillation and ν is the kinematic viscosity, is a very important parameter in pulsatile flow. It measures the extent to which an oscillatory velocity profile differs from the steady Poiseuille profile. It is sometimes viewed as an unsteady Reynolds number because it compares the significance of inertial and viscous effects over one oscillation period.

5.3.2 Small Womersley Number

When $\alpha \ll 1$, viscous forces dominate and the flow is described to be quasi-steady. In addition, the velocity profile remains parabolic at all times with the instantaneous flow rate determined by the instantaneous pressure gradient. Although the pressure gradient varies in time, the flow still behaves very similar to the steady Poiseuille flow. Since viscous forces dominate, inertial effects may be neglected, meaning that the fluid responds almost immediately to changes in pressure hence there is no lag between the applied pressure gradient and the resulting flow.

This type of behaviour is typically seen in the microcirculation. In small vessels, $\alpha = 0.08$ in $100\ \mu\text{m}$ vessels and approximately $\alpha = 0.005$ in capillaries [6]. Under these conditions, the motion is determined almost entirely by the balance between pressure and viscous forces, with inertial forces becoming negligible.

5.3.3 Intermediate Womersley Number

As α increases, inertial effects begin to influence the flow more strongly, leading the velocity profile to become more distorted relative to the Poiseuille profile. Furthermore, the instantaneous flow rate is no longer determined solely by the instantaneous pressure gradient. Instead, the flow begins to lag behind the applied pressure gradient. This lag indicates that the velocity field is no longer in phase with the pressure gradient, due to the increasing influence of fluid inertia. For values of α in the approximate range 1 to 3 [6], this phase lag becomes more noticeable.

5.3.4 Large Womersley Number

For even larger values of α ($\alpha > 4$) [6], inertial forces dominate more strongly. The phase lag between pressure and flow increases and the peak flow rate becomes smaller than predicted from the peak pressure gradient. This shows that increasing α shifts the flow away from a Poiseuille-like behaviour. At very high frequencies, the oscillatory motion becomes more concentrated in a thin region near the wall, while the motion in the core is much smaller. This means that viscous forces are more dominant at the wall, whereas inertial forces are more dominant in the core.

This behaviour is common in large arteries, where pulsatile effects are far more significant than in the microcirculation. Therefore, the Womersley number provides a useful explanation for why flow in large arteries cannot generally be approximated by the steady

Hagen-Poiseuille model.

5.4 Limitations of the Womersley Model

This chapter took the oscillatory pressure gradient into account, developing a more realistic model for blood flow in larger arteries. Although the full velocity profile was not derived, the Womersley number was obtained instead. This parameter describes how the flow responds to a time-dependent pressure gradient when either viscous or inertial forces dominate.

However, the Womersley model still assumes a uniform cylindrical geometry and rigid walls. However, it does not account for the complex rheology of blood nor the viscoelastic properties of the vessel walls. The next chapter discusses further work, including viscoelasticity, time-dependent boundary conditions from elastic vessel walls and the effects of less deformable RBCs in sickle cell disease.

6 Further Work

6.1 Viscoelasticity of Blood and Vessel Walls

A viscoelastic material is one that possesses the properties of both elastic and viscous behaviour simultaneously. This requires a combination of Newton's law (for viscous behaviour) and Hooke's law (for elastic behaviour), where stress is proportional to strain to describe its overall behaviour. Given that blood vessels contain elastin and collagen, these contribute to the elastic aspect while the presence of smooth cells contributes to the viscous response of the wall. One key feature of elastic behaviour is memory which means that the material can store energy and return to its original shape after the applied stress is removed.

Previously, blood was modelled as Newtonian, where the wall shear stress was assumed to be directly proportional to shear rate, such that $\tau = \mu\dot{\gamma}$ where γ is the shear strain ($\dot{\gamma} = \frac{d\gamma}{dt}$), or equivalently $\tau = \mu\frac{\partial w}{\partial n}$. However, with the introduction of elastic effects, the stress is no longer determined solely by the rate of deformation but it also depends on the deformation itself. This leads to a viscoelastic constitutive relation of the form

$$\tau = G\gamma + \mu\dot{\gamma} \tag{6.1}$$

which combines the elastic response (through shear strain γ) and the viscous response (through the shear rate $\dot{\gamma}$) in a single model.

As a result, the wall shear stress depends not only on the velocity gradient, but also on the history of deformation by introducing a time-dependent aspect absent in Newtonian models, where G is the elastic modulus and μ is the viscosity.

6.2 Extension of Womersley Flow

The Womersley flow captures the unsteady pulsatile nature of blood flow which arises from the periodic pumping of the heart. A key limitation is the assumption of rigid walls, which implies that the flow remains strongly pulsatile throughout the arteries.

In reality, arterial walls are elastic and deform in response to changes in pressure. When blood is ejected into the arteries, the vessel expands to accommodate the increased pressure and volume. As the pressure decreases, the vessel recoils back towards its original shape. This recoil helps maintain forward flow during diastole, smoothing pulsatile behaviour and making the flow more continuous.

6.2.1 The Moving Radius and Time-Dependent Boundary Conditions

To model the effect of elastic walls, the vessel radius can be allowed to vary in both space and time so that $R = R(z, t)$. This reflects the propagation of a wave along the vessel as pressure changes travel through the arterial systems.

A simple model is to assume that the radius takes the form

$$R(z, t) = R_0 + \varepsilon \sin(kz - \omega t), \quad (6.2)$$

where R_0 is the mean radius of the vessel, ε is the amplitude of the oscillation, k is the wave number, ω is the angular frequency and $t \in \mathbb{R}_{\geq 0}$.

The transition from a rigid wall to a moving, elastic wall also leads to a change in the boundary conditions from static to dynamic. When the vessel wall moves, it possesses its own velocity, which must be incorporated into the boundary conditions. As a result, the no-slip condition is no longer applied at a fixed boundary and instead it is applied at a boundary that varies in time. This leads to time-dependent boundary conditions.

6.2.2 Example of Time-Dependent Boundary Conditions

Given Equation (6.2), the radius at a fixed location $z = 0$ is given by

$$R(z, t) = R_0 + \varepsilon \sin(-\omega t) \quad (6.3)$$

Then the velocity of the wall at $z = 0$ is given by

$$\frac{dR}{dt} = \varepsilon \omega \cos(\omega t). \quad (6.4)$$

Using an aortic diameter of approximately 2.5 cm, the mean radius is $R_0 = 1.25$ cm. Experimental studies have reported oscillations in arterial diameter of approximately 3%–4% of the mean value [13], so the amplitude is given by

$$\varepsilon = 0.03R_0 = 0.000375 \text{ m}$$

Taking the typical heart rate of 60 bpm gives a frequency of $f = 1$ Hz, hence $\omega = 2\pi f = 2\pi \text{ rad s}^{-1}$ and the velocity of the wall is given by

$$\frac{dR}{dt} = (0.000375)(2\pi) \cos(2\pi t) \quad (6.5)$$

Therefore, the maximum velocity is 0.00236 m s^{-1} . Given the no-slip condition, $\mathbf{u}_{wall} = \mathbf{u}_{fluid}$, this is also the velocity of blood and the boundary.

6.3 Less Deformable Blood Cells: Sickle Cell Disease

Blood has been modelled as a Newtonian fluid with constant velocity. The RBCs in blood are primarily responsible for the viscous behaviour of the suspension. These cells are deformable allowing them to change shape under the action of the flow and move more easily with the surrounding fluid.

However, when the cells are less deformable and more rigid, they resist this motion and disrupt the flow. This forces the surrounding fluid to shear more strongly, causing an increase in viscosity and hence an increase in the overall resistance.

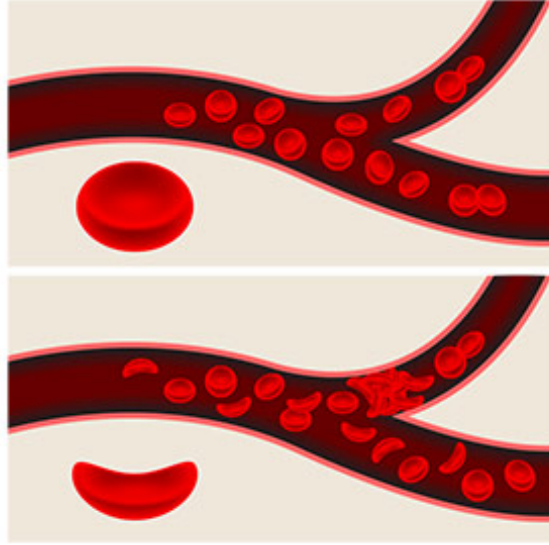


Figure 6.1: Deformable healthy RBCs (top) vs less deformable sickle cells (bottom) [4].

A real-life example of this is sickle cell anaemia, where the shape of RBCs take on a *sickle* shape causing them to be less deformable. In addition, immature RBCs, known as reticulocytes, are more likely to adhere to the vessel walls, which can further disrupt the flow, as discussed in Chapter 3.

7 Conclusion

This project investigated the use of the continuity equation and momentum equations to describe the behaviour of blood in blood vessels across a range of geometric structures. Starting with a simple model, the governing equations were used to describe flow in a

rigid cylindrical tube, simulating large arteries. The model was then extended to consider more complex cases including the presence of blockages, flow in capillaries and oscillating pressure gradients before exploring further extensions.

The modelling showed how wall shear stress, velocity and pressure responded to changes in vessel structure. In a cylindrical tube, a reduction in a radius due to a blockage leads to an increase in velocity, $u \propto R^{-2}$, an increase in wall shear stress, $\tau_w \propto R^{-3}$ and an increase in pressure gradient, meaning the pressure decreases more rapidly in the constricted region.

Flow in capillaries was examined through creeping flow, where viscous forces dominate and inertial effects are negligible. This justifies the use of lubrication theory and highlights how flow behaviour differs significantly from that in larger vessels.

Introducing an oscillating pressure gradient led to the Womersley model, characterised by the Womersley number. This measures how the flow deviates from the Hagen-Poiseuille behaviour. For small values of the Womersley, the flow remains close to Poiseuille flow. For intermediate values, inertial effects become more significant and a phase lag is introduced between the pressure gradient and velocity. For large values of the Womersley number, this lag becomes more pronounced, providing a more accurate description of flow in arteries.

Finally, further work was explored to understand the limitations of the models used. The assumption of Newtonian behaviour neglects the complex rheology of blood. However, the introduction of viscoelasticity highlights how both the blood and vessel walls exhibit combined viscous and elastic behaviour. In addition, the assumption of rigid vessel walls limit the realism of the model as in reality, the radius may vary with both space and time. This led to the introduction of time-dependent boundary conditions. Lastly, considering less deformable blood, such as in sickle cell conditions, shows how increasing rigidity affects viscosity. Incorporating these effects would therefore provide a more realistic representation of the flow.

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